

Data-Quality Improvements and Applications of Long-Term Monitoring of Ionospheric Anomalies for GBAS

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ABSTRACT

An automated Long-Term Ionospheric Anomaly Monitoring (LTIAM) software package has been developed to support continuous ionospheric monitoring for the U.S. Local Area Augmentation System (LAAS) developed by the U.S. Federal Aviation Administration (FAA) in the Conterminous U.S. (CONUS). Continuous monitoring is needed to confirm the long-term validity of the ionospheric anomaly threat model [1,3] and update it if necessary. This is of particular importance over the next few years, as the intensity of solar storms is expected to peak in 2013-15. Prior work [2,4] has demonstrated that the LTIAM not only identifies extremely large ionospheric gradients but also supplies reliable ionospheric gradient statistics under all conditions. This tool can also be utilized to build ionosphere threat models for all regions where Ground-Based Augmentation Systems (GBAS) will be fielded in the future.

The LTIAM software enables the post-processing of data continuously collected by GPS reference station networks. Ionospheric gradients over short-baseline distances of 5 – 40 km can be observed using data collected from the Continuously Operating Reference Stations (CORS) network, which has over 1800 stations as of 2011 compared to about 400 stations prior to 2004. However, as the total number of stations increases, the number of stations with poor GPS data quality also increases. This is caused by the differing types of CORS receivers and antennas fielded by multiple organizations in various environments; some good, some not-so-good. The use of poor-quality data degrades the accuracy of ionospheric delay estimates and produces too many faulty anomaly candidates. Thus, station selection criteria need to be

defined to reduce processing time in both automated procedures and manual analysis and validation while maximizing the benefit of this tool.

1.0. INTRODUCTION

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2.0. LTIAM OVERVIEW

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3.0. POOR CORS DATA QUALITY

This section investigates the effect of GPS data collected from stations with poor GPS data quality on the results of the LTIAM tool. Figure 1 shows the results from LTIAM with a threshold of 200 mm/km on a nominal day, 26 May 2012, during which geomagnetic storm condition was very quiet. On this particular day, it is expected that any severely large ionospheric gradients are not observed. However, the LTIAM tool returned many ionospheric anomalies because of the bad GPS data used. All 92 faulty threat candidates, marked with blue diamonds in ionospheric anomaly threat space, should be manually validated to confirm that these points are not real anomalies. If stations with poor GPS data quality are effectively removed by a methodology presented in this paper, the most of faulty candidates can be removed as shown in Figure 2. Therefore, we can save time and effort spending to manually validate the faulty candidates.

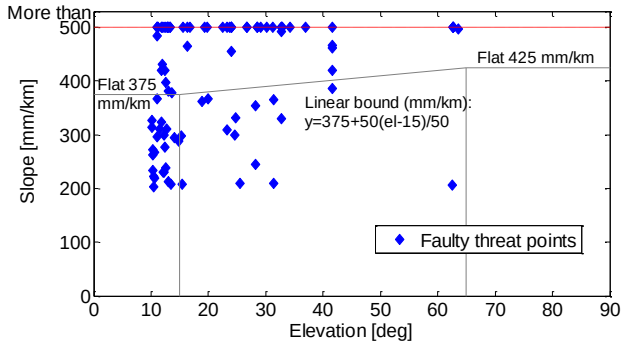


Figure 1. 92 Faulty Anomaly Candidates Populated on Ionospheric Anomaly Threat Space Before Removing Stations with Poor GPS Data Quality on 26 May 2012.

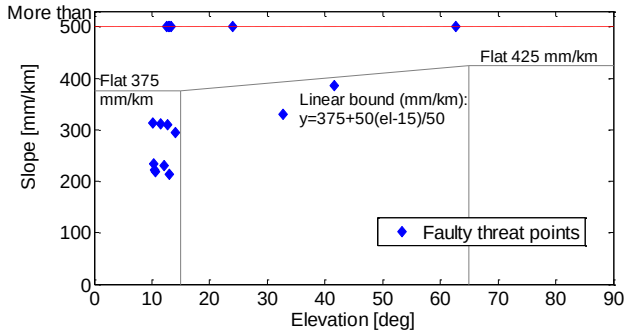


Figure 2. 11 Faulty Anomaly Candidates Populated on Ionospheric Anomaly Threat Space After Removing Stations with Poor GPS Data Quality on 26 May 2012.

Figure 3 shows the slant ionospheric delays observed from two nearby stations OKEE and AVCA while they tracked PRN 22. OKEE is a good example of a station with poor GPS data quality. From many fragmentations of ionospheric delay estimates of OKEE (red), it is evident that carrier-phase measurements are corrupted by numerous cycle slips, short arcs, and outliers. OKEE (red) also held a small amount of data compared to AVCA (blue). Figure 4 shows the ionospheric spatial gradients between OKEE and AVCA. The leveling errors due to the short arcs are observable at each end of the curve, and the large gradients due to the excessive cycle slips are evident in the center of the curve. This example well illustrates that the poor data quality degrades the accuracy of ionospheric delay estimation and produce extremely large ionospheric gradients which are not real.

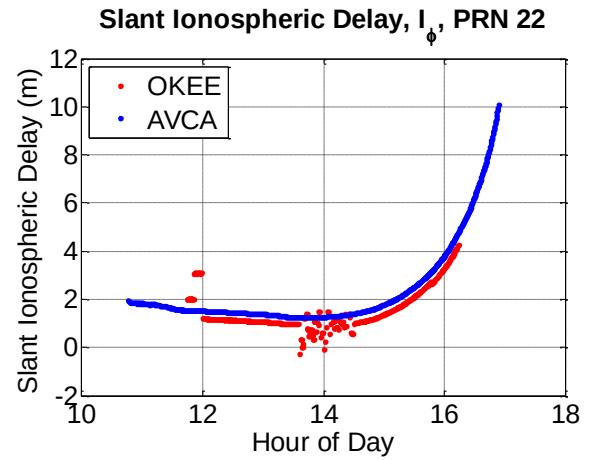


Figure 3. Dual-frequency Slant Ionospheric Delay Estimates for Stations OKEE (Poor Quality Data) and AVCA (Good Quality Data).

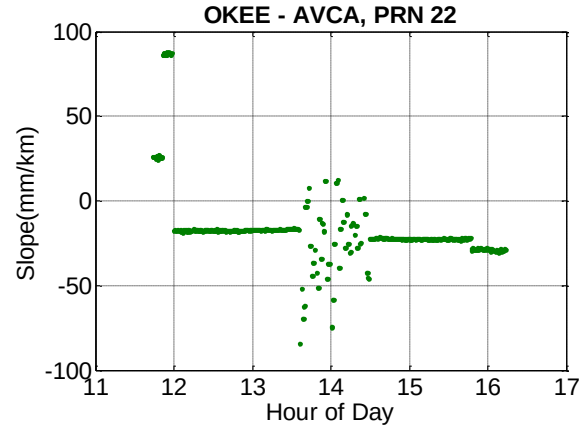


Figure 4. Ionospheric Gradient Estimates Corrupted by the Poor Quality Data of Station OKEE.

4.0. DATA ANALYSIS METHODOLOGY

The detailed methodology to select the station for LTIAM is composed of the determining GPS data quality information, the determining criteria to select the station and geometry check as shown in 오류! 참조 원본을 찾을 수 없습니다.. First, the data quality information is obtained by processing the RINEX file collected at each station. Then, the stations with poor data quality that need to be removed are selected based on whether the quality parameters of the station exceed the established thresholds. These selected stations are ranked based on the data quality and data sampling rate. Then we examine the degree of geographical contribution of selected stations in the order of this rank. Finally, we put back stations which increase geographical coverage despite poor data quality. The details of each step are described in the following subsections.

4.1. DATA QUALITY DETERMINATION

TBC

4.2. GEOGRAPHICAL DENSITY CONDITION

5.0. RESULTS OF CORS STATION REMOVAL

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6.0. UPDATED THREAT ANALYSIS FROM PAST STORM DATA

As explained above, the LTIAM is primarily intended to examine new CORS and IGS station data to detect recent and future anomalous ionospheric gradients. It also has the capability to re-examine past ionospheric storm data and discover new properties of it, as shown in [22]. This section continues the analysis in [22] to better understand the distribution of spatial gradients under anomalous ionospheric conditions. It then uses this analysis as a template for estimating anomalous gradient probabilities from future ionospheric data.

6.1. EXTENDED ANALYSIS OF 2000-2005 RESULTS

Table A (from [22]) shows the 10 days of known ionospheric storm activity in CONUS from 2000 to 2005 that were manually analyzed prior to the existence of LTIAM [2,3] and re-analyzed with LTIAM [22] to develop the current GBAS ionospheric threat model for CONUS. Figure A is one of the key results from [22] and shows the observed and validated anomalous ionospheric gradients over these 10 days. This includes gradients previously discovered from manual analysis and later re-confirmed by LTIAM (green triangles) and those first discovered more recently by LTIAM (blue diamonds). All of the observations shown in this figure were derived from the four days highlighted in yellow in Table 1. These four days, and especially 11/20/03, were caused by an especially strong coronal mass ejection from the Sun and represent the most severe storm days known to have occurred since GPS data has become available [2,3].

Table A. Storm Dates Analyzed to Develop CONUS Ionospheric Threat Model [22]

Day (UT mm/dd/yy)	Kp	Dst
04/06/00	8.3	-287
04/07/00	8.7	-288
07/15/00	9.0	-289
07/16/00	7.7	-301
09/07/02	7.3	-177
10/29/03	9.0	-350
10/30/03	9.0	-383
10/31/03	8.3	-307
11/20/03	8.7	-422
07/17/04	6.0	-76

The CONUS threat model limits shown in Figure A were chosen to bound the largest observed events. For this reason, previous data searches focused on identifying and validating the largest apparent gradients in the dataset. For automated LTIAM data analysis, this was done by limiting the outputs to apparent gradients above 200 mm/km (as measured by LTIAM from raw L1/L2 data). While some of the resulting validated gradients shown in Figure A are lower than this threshold, this is because validated gradient values are those that can be confirmed by comparing L1/L2 estimates with those from L1 code-minus-carrier analysis. When a discrepancy exists for an observation that appears to be valid, the minimum gradient that can be confirmed to be valid is reported, and many of these are under the original LTIAM reporting threshold. Therefore, the distribution of gradients in Figure A is roughly uniform (equally-distributed) between the minimum that represents anomalous conditions (25 – 50 mm/km) and the maximum of just over 400 mm/km. This does not appear to be a fair reflection of all anomalous conditions, in which we would expect smaller gradients to be much more frequent than larger ones.

To examine this further, the LTIAM was used to re-evaluate the 10 storm days in Table A with a lower threshold of 50 mm/km and an upper threshold of 200 mm/km. In other words, only gradients that pass the internal LTIAM checks and have estimated magnitudes between 50 and 200 mm/km were reported. This avoids any overlap between new observations and those made previously with a minimum threshold of 200 mm/km. Because of the large number of gradients discovered in this range, manual validation is not practical; thus the reported gradients are the "unvalidated" values from L1/L2 analysis and will tend to be larger than the equivalent "validated" ones for the reasons mentioned above.

Figure B shows the results in the same format as Figure A: gradient (or "Slope") vs. satellite elevation angle. A total of 2929 points were discovered between 50 and 200 mm/km. What is evident is that the number of gradients

of smaller magnitudes greatly exceeds the number above 200 mm/km. In addition, between 50 and 200 mm/km, it is clear that smaller gradients are more likely. Figure C emphasizes this by showing the cumulative distribution of gradients between 50 and 200 mm/km (in other words, the x-axis probability of falling below the gradient indicated on the y-axis). Figure C shows that the median (50th percentile) of these gradients is 82.8 mm/km, while the midpoint of the range from 50 to 200 mm/km is 125 mm/km. Similarly, the 90th percentile is 149.1 mm/km, while the 90th percentile of the range is 185 mm/km.

The key unknown in evaluating the distribution of anomalous gradients is the fraction of the 2929 observations that are valid; i.e., they represent actual ionospheric gradients. We know from experience with the LTIAM is that lower apparent gradients are more likely to be valid than larger ones. The reason for this is that actual ionospheric gradients are limited by physics, while "false" anomalies due to receiver or database errors are not limited in this way and can take almost any size. Therefore, it is almost certain that the percentage of valid observations from 50 to 200 mm/km is higher than the same percentage above 200 mm/km. This percentage is about 30%, based on 73 validated measurements above 200 mm/km out of 243 outputs generated by LTIAM [22]. Note that Figure A shows a total of 99 validated measurements, but 26 of these resulted in validated events below 200 mm/km, and these are counted as being in the 50 – 200 mm/km range. As expected, a slightly smaller percentage applies for an LTIAM threshold of 300 mm/km (13 of 53, or about 24.5%).

If we use 30% as a lower bound on the percentage of valid observations between 50 and 200 mm/km, the resulting estimate is $2929 \times 0.3 = 878.7$. Adding the 26 previously validated observations in this range gives a total of 904.7, or about 905 valid observations. The 73 validated observations above 200 mm/km thus represent a fraction of $73/(905+73)$, or about 7.5% of the total set of (estimated) validated observations. Above 300 mm/km, the ratio is $13/(905+73)$, or about 1.3% of the total set. If we instead assume as an upper bound that all 2929 measurements between 50 and 200 mm/km are valid, the ratio of validated observations above 200 mm/km would drop to $73/(2929+26+73)$, or about 2.4%, while the ratio above 300 mm/km would drop to $13/(2929+26+73)$, or about 0.4%.

These bounds show that, with very high confidence, the fraction of anomalous gradients above 200 mm/km in the 2000-2005 CONUS ionospheric storm data set is very small: below 10% at least, and very likely below 5%. Furthermore, since the distribution of LTIAM measurements between 50 and 200 mm/km is weighted toward the lower end, and lower gradients are more likely to be valid than higher ones, it is safe to conclude that the

distribution of gradients throughout the CONUS threat space is heavily weighted toward the lower end of the gradient range. While GBAS ground stations must conservatively protect the entire threat space, including the highest possible gradients, this knowledge is not directly applicable to the ionospheric threat mitigations described in [1,4,5]. However, it helps us better understand ionospheric behavior relevant to GBAS during anomalous conditions, and it should lead to improved and less-conservative mitigation strategies in the future.

6.2. UPDATING PROBABILITY ESTIMATES WITH FUTURE DATA

The primary objective of LTIAM is to continually improve our understanding of both nominal and anomalous ionospheric gradient behavior. The re-analysis of past data in Section 6.1 shows an example of what can be learned. Going forward, the processing power of the LTIAM software allows automated analysis of all days in a particular region or a focus on particular days that appear anomalous based on external information such as ionospheric weather parameters [6,22]. The information collected over time from these results will allow us to update the probabilities estimated in Section 6.1 and gain a better idea of the "prior probability" of extreme ionospheric gradients (e.g., over 200 mm/km) (a) over all ionospheric conditions; and (b) conditional on the knowledge that anomalous conditions are present. It is important to distinguish these two results. If LTIAM processing of all data in a particular region is carried out, the distinction between "nominal" and "anomalous behavior" can be determined after the fact based on both the observed gradients and external space weather information. In this case, probability estimates under all conditions and under anomalous conditions can be computed separately. If LTIAM processing is only carried out for days thought to be anomalous based on external information, only the latter estimate can be made.

Updating the results in Section 6.1 is mostly a process of repeating the same analysis procedure with future data while including data already analyzed from the past. The desired result is a data-driven distribution of anomalous gradients, meaning gradients above 25 or 50 mm/km. The lower number represents the lower limit of the anomalous gradient space in the CONUS threat model, but gradients at this level are not threatening to GBAS. Therefore, it may be more practical to limit LTIAM searches to 50 mm/km or more, as done in this paper. While anomalous gradients of any magnitude are rare, continual monitoring will occasionally discover them and allow new points to be added to the observations made to date. Thus, our knowledge of the distribution of anomalous gradients will grow slowly with time.

The rarity of severely anomalous gradients makes estimating their prior probability difficult. The model proposed in [23] uses the information available in early 2006 to estimate the probability of days with extremely anomalous ionospheric behavior, defined as days with gradients above 200 mm/km that could threaten GBAS (as shown in Table A, all such days in the 2000-2005 database had Kp indices greater than 8.0). This probability was estimated as a mean of 0.00196 and a 60th percentile upper bound of 0.00257.

The ability of LTIAM to process all days (or at least all days with significant ionospheric activity) allows us to improve upon the precision and usefulness of these numbers. Given a set of days analyzed (either “all days” or “days of significant activity”), the first probability to be computed is the fraction of days with any anomalous gradients, defined as either above 25 or above 50 mm/km. Combining this result with the distribution of gradients above 25 or 50 mm/km allows the computation of the probability of observing gradients above any particular magnitude (e.g., 200 mm/km) per unit time. While “days” was the time interval used in [23], LTIAM outputs can and should be grouped into smaller intervals, likely “hours,” to better reflect changing ionospheric conditions over the 24-hour “daily” cycle as well as the relatively short duration of anomalous gradients affecting particular areas.

7.0. SUMMARY

This paper provides an overview of the LTIAM software processing tool and demonstrates how it is used to identify potential ionospheric spatial anomalies from raw data collected by existing networks of CORS and IGS stations. Although LTIAM does its own screening of the raw data, significant ionospheric anomalies are rare; thus most of the results from LTIAM are “false” anomalies created by receiver or database errors. The vast majority of “false” anomalies come from relatively small subsets of CORS and IGS stations with poor data quality. The data quality evaluation methodology developed in this paper successfully reduced the number of false anomalies by almost 90% by removing only 16% of the CORS stations in CONUS. This was achieved while retaining stations with marginal data quality in geographically key locations. The end result is that the number of outputs requiring manual validation is greatly reduced.

This paper also re-examines the database of known ionospheric storm events in CONUS from 2000 to 2005 to estimate the distribution of anomalous gradient magnitudes. Examining all gradients estimated to be greater than 50 mm/km by the LTIAM shows that the validated events with gradients above 200 mm/km are greatly outnumbered by events with gradients from 50 to

100 mm/km. The degree to which gradients above 200 mm/km are unusual depends upon the percentage of valid events from 50 to 200 mm/km, which is almost certainly larger than the approximately 30% that applies to anomalies above 200 mm/km in the same dataset. A simplified method is proposed to update this probability and estimate the overall probability of anomalous ionospheric conditions as more data is collected over time.

ACKNOWLEDGMENTS

The authors thank the Federal Aviation Administration (FAA) Local Area Augmentation System (LAAS) Program Office, of whom John Warburton and Jason Burns were particularly helpful. We also would like to thank Attila Komjathy of the Jet Propulsion Laboratory, and Per Enge, Seebany Datta-Barua, Todd Walter, and Juan Blanch of Stanford for their support of this work.

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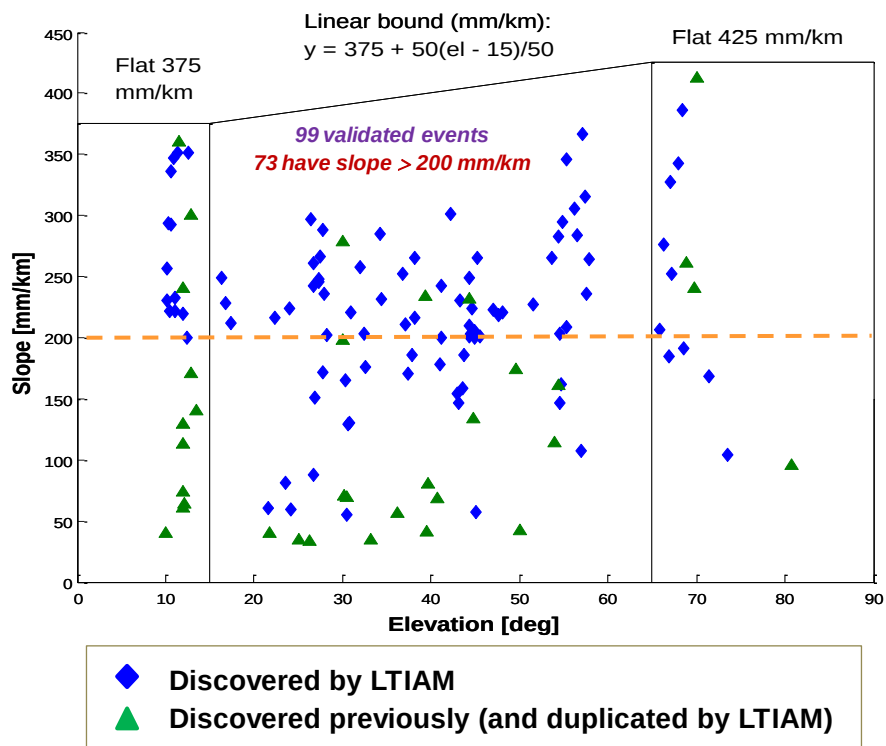


Figure A. CONUS Ionospheric Threat Model with Observed Spatial Gradients (LTIAM Threshold = 200 mm/km)

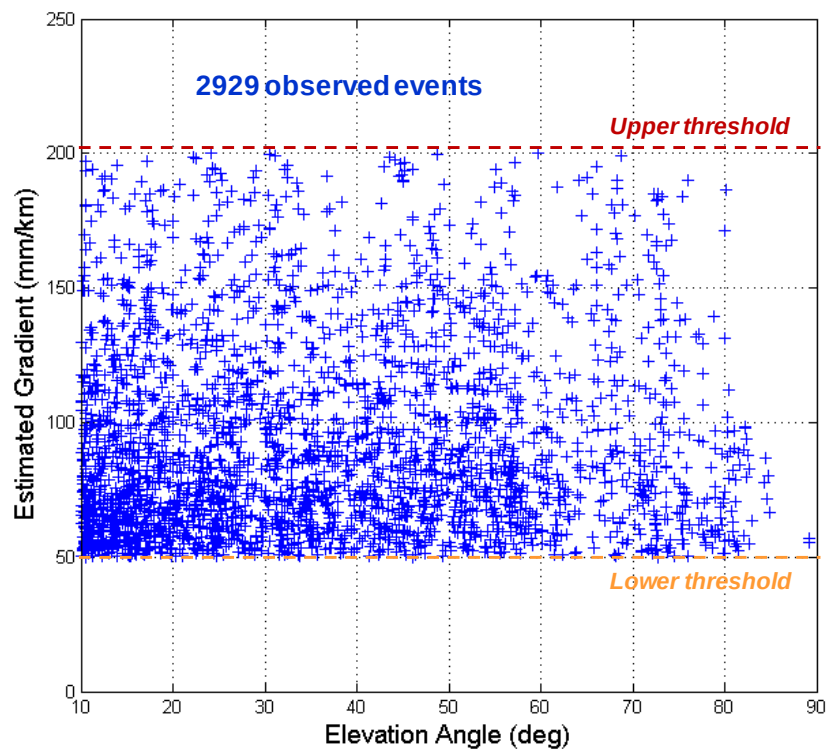


Figure B. CONUS Ionospheric Gradients from LTIAM (50 to 200 mm/km)

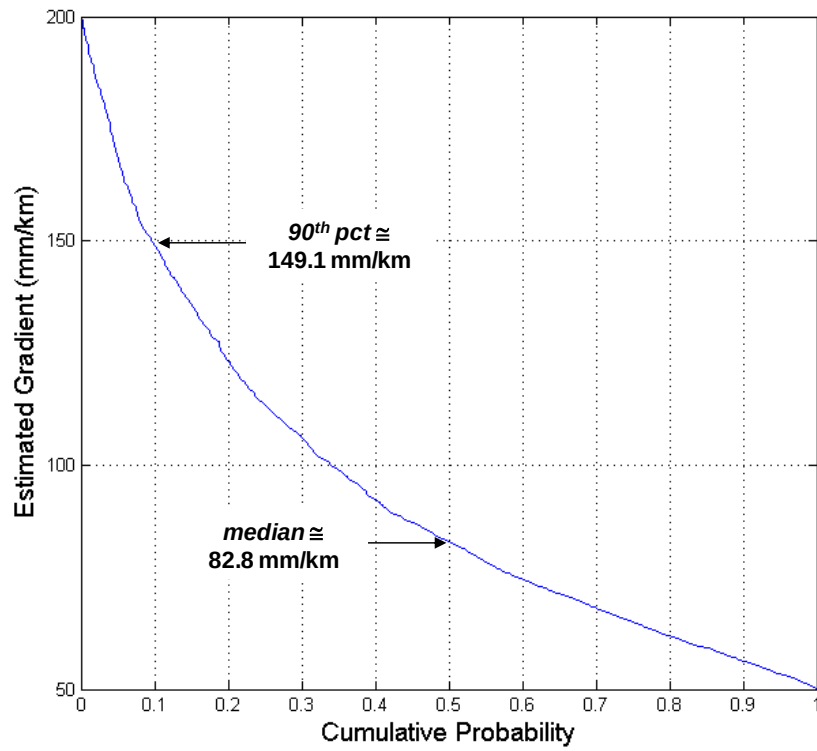


Figure C. Cumulative Distribution of CONUS Ionospheric Gradients from LTIAM (50 to 200 mm/km)